Multimodal Emotion Recognition based on Convolutional Neural Networks and Long Short-Term Memory Networks

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***Abstract*—Due to the rapid advancements in big data and high-performance computing technologies, the application of deep learning models in the fields of computer vision and signal image processing has become increasingly widespread. To address the low efficiency and accuracy in the recognition of multimodal emotional EEG signals, this study integrates the latest deep learning network models, leveraging the advantages of convolutional neural networks (CNNs) in EEG feature extraction and the strengths of long short-term memory (LSTM) networks in processing dynamic temporal information. Therefore, this study employs CNN models to extract features from multimodal EEG signals, followed by feature vector fusion using LSTM, and classification using softmax. The network model constructed in this study achieved an accuracy of 92% in the three-class emotion recognition. Compared with traditional classification methods, the CNN-LSTM network model exhibits higher recognition efficiency due to its combined advantages in feature extraction and dynamic temporal information processing. Compared to using CNN or LSTM models alone, this model also shows a significant improvement in recognition accuracy.**

Keywords—Deep learning, multimodal signals, EEG emotion recognition, CNN-LSTM

I. INTRODUCTION

Over the past few decades, many methods for emotion recognition based on physiological features have been proposed. Those based on subconscious external features primarily include facial expressions, voice, and posture[1-2]. These signals are relatively easy to collect. However, when participants are unwilling to express their true emotions during experiments, they can easily disguise their feelings, which can undermine the validity and accuracy of emotion recognition models. At this stage, emotion recognition relies on internal bodily signals, primarily including electroencephalogram (EEG), electrocardio-gram (ECG), and other physiological signals.[3]. These physiological signals are objectively more reliable for emotion recognition. EEG signals have emerged as the primary data source in emotion recognition research due to their objectivity, authenticity, inability to be masked, real-time variability of data, and ease of acquisition[4-5].

Emotion recognition, a fundamental task in achieving comprehensive artificial intelligence, has emerged as a global research focus in fields such as safe driving and health monitoring[6-8]. Currently, traditional machine learning methods achieve a degree of effectiveness in classifying EEG signals; Laura et al. employed naive Bayes, support vector machines with different kernels, random forests, and artificial neural networks to classify emotions in the EEG dataset[9]. However, there is a need to improve both feature extraction efficiency and classification accuracy. With the application of deep learning in the field of EEG emotion recognition, deep neural network models can handle preprocessing of EEG signals, feature extraction from signal graphs, and classification of multidimensional features. [10].

EEG emotion recognition using deep learning algorithms achieves higher-dimensional features compared to manually designed features and reduces recognition time. Specifically, the widespread adoption of algorithms such as CNN, LSTM, Recurrent Neural Networks (RNN) has greatly enhanced the efficiency and accuracy of emotion recognition. For example, Wu et al. captured spatial information in signals using three distinct statistical measures. They carefully explored the complementary information across various CNN hidden layers and enh-anced feature representation capabilities through a multi-feature fusion module, achieving a binary classification accura-cy of 90.57%[11]. Kim et al. achieved an accuracy of 82.86% in classifying emotions into three categories using LSTM[12].

Building on this, this paper will use CNN to extract multi-channel feature vectors from EEG data, fuse these feature vectors with LSTM, and integrate them with dynamic time series information to achieve emotion classification. This approach is outlined in three key points:

(1) Designing a 1D CNN to exploit weight sharing and local perception characteristics, which extracts multi-channel

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features from 62 channels of raw EEG data to form multi-channel feature vectors.

(2) Developing an LSTM model capable of effectively handling time-series data by merging the output multi-channel feature vectors for final classification recognition.

(3) Comparing the model proposed in this study with independent models and traditional machine learning classification approaches to validate its superior performance in multimodal emotion recognition.

II. Deep Learning Model Design

Given that convolutional neural networks and long short-term memory networks each excel in feature extraction and dynamic temporal information processing, the deep learning model designed in this paper, which integrates both network models, is depicted in Figure 1. Firstly, a one dimensional CNN automatically extracts emotional features of EEG signals from each channel. Then, LSTM networks model the sequential information to extract fused emotional features from 62 channels. Finally, a Softmax classifier categorizes the fused emotional features across multiple channels into three categories: positive, neutral, and negative emotions[13].

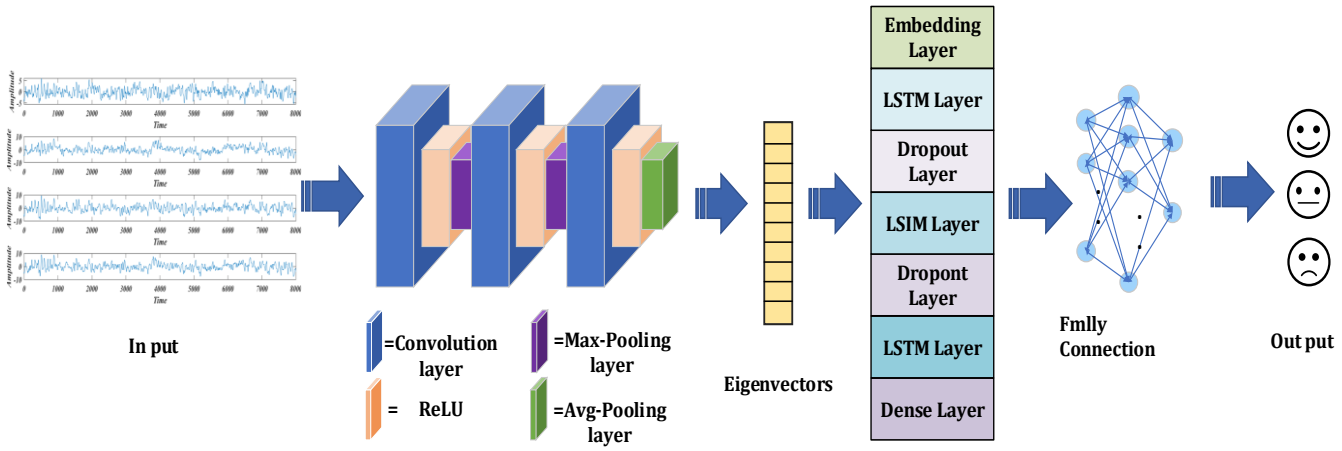


Figure 1. Multimodal emotion classification process based on the CNN-LSTM model.

A. CNN model design

In addressing the classification of multi-channel data, CNN models, in comparison to fully connected neural networks, can balance local content and distribute weights, significantly reducing model complexity[14]. The structure of the 1D CNN is depicted in Figure 2, comprising three convolutional layers, two max pooling layers, and one average pooling layer.

The convolutional layer convolves local receptive fields from the previous layer. Each convolutional layer uses 1x5 convolutional kernels with a stride of 1. ReLU activation function is used after the convolutional layer to introduce non-linearity. It causes some neurons in the network to output zero after training, achieving moderate sparsity. This speeds up the convergence of the network and reduces interdependencies among parameters. This avoids overfitting issues in the model. Thus, it enhances the generalization capability of the model. The output features of the average pooling layer are high-level features from a single channel. Through multiple stages of training, features are extracted progressively and accurately.

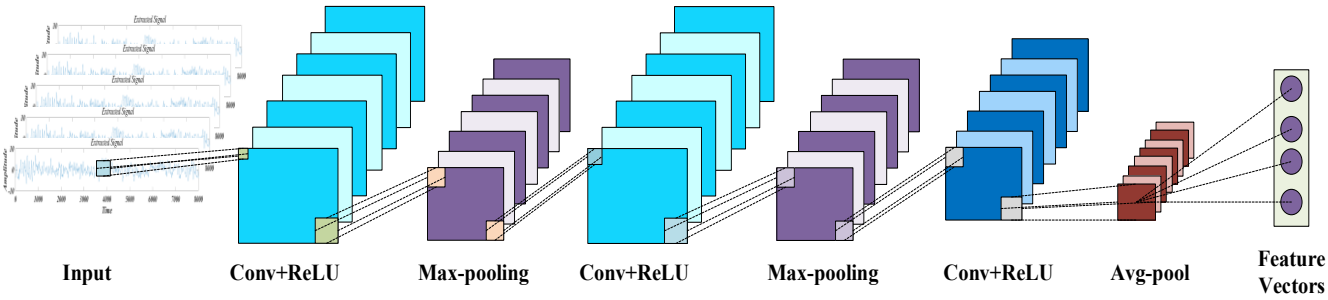


Figure 2. Process of CNN model extracting EEG feature vectors

B. LSTM Network model design

Since LSTM networks are a type of RNN, they are particularly suitable for modeling time-series data. The primary benefit of LSTM networks over conventional RNNs is their capacity to identify long-term dependencies in data by employing memory cells, which have the capacity to retain information over time. Therefore, the obtained multi-channel EEG signal feature vectors are input into LSTM. Multi-channel fused emotion features are extracted from emotional features of each channel. As shown in Figure 1, the LSTM network architecture designed in this study consists of three LSTM layers, each followed by a fully connected layer. The LSTM layers are in charge of learning associations between various time points

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and capturing the temporal dynamics of EEG data. The specific units of LSTM are shown in Figure 3. Information is updated via input, forget, and output gates. Which data from the previous time step should be kept or discarded is decided by the forget gate[15]. The input gate is used to update the current state information. Which data should be forwarded to the following time step is decided by the output gate.

In the first step, the activation function sigmoid, input weights W , and input biases b are used, along with the hidden state from the previous unit h_{t-1} and the current multi-feature vector input x_t , to calculate the forget gate, input gate, and output gate. Simultaneously, the activation function tanh is applied to obtain the candidate memory cell $C_t^{'}$.

$$f_t = sigmoid\left(W_f \times [h_{t-1}, x_t] + b_f\right) \quad (1)$$

$$i_t = sigmoid\left(W_i \times [h_{t-1}, x_t] + b_i\right) \quad (2)$$

$$o_t = sigmoid\left(W_o \times [h_{t-1}, x_t] + b_o\right) \quad (3)$$

$$C_t^{'} = \tanh\left(W_c \times [h_{t-1}, x_t] + b_c\right) \quad (4)$$

The second step involves using the forget gate f_t to determine how much of the past memory cell C_{t-1} should be retained, while the input gate i_t controls the amount of new data from $C_t^{'}$ that is used to update the memory cell C_t . The process is as follows.

$$C_t = C_{t-1} \times f_t + i_t \times C_t^{'} \quad (5)$$

The final step is to determine the model's output by using the output gate o_t and the activated memory cell C_t , which is activated by the tanh function, to determine the new hidden state ht.

$$h_t = o_t \times \tanh\left(C_t\right) \quad (6)$$

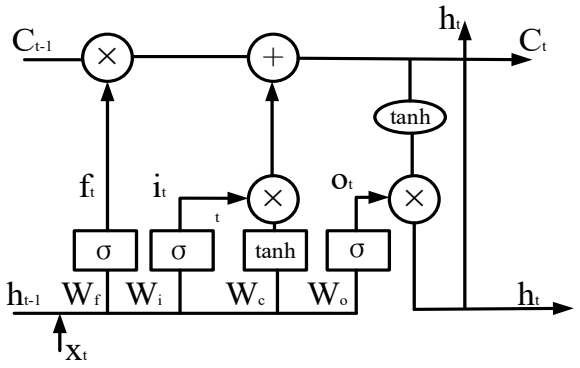


Figure 3. LSTM structure diagram

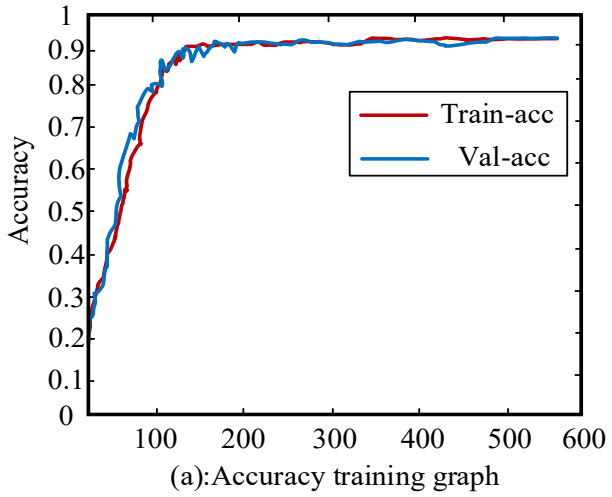
C. Data parameter adjustment

This study uses the SEED dataset to assess the effectiveness of the CNN-LSTM model in multimodal EEG emotion recognition, as this dataset provides a more accurate representation of various human emotions[16]. After preprocessing, the dataset is structured as a four-dimensional matrix of subjects × trials × channels × samp-ling points. It was reshaped by reducing dimensions to a three-dimensional matrix of trials × channels × sampling points. Classification accuracy was obtained by cross-trial classification to build the dataset. 70% of the entire dataset was chosen for training and 30% for testing.

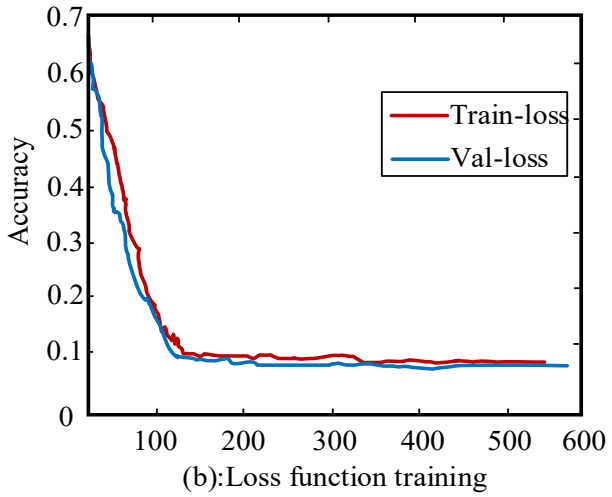
III. Performance and Classification Results of Deep Learning Models

A. Performance of CNN-LSTM model

After merging the extracted multi-channel feature vectors, these vectors were trained in the model to produce the corresponding accuracy and loss iteration curves. Figure 4 illustrates the changes in model accuracy and the loss function. In this training, the blue line represents the training data, while the red line represents the validation data. Accuracy improved significantly during the first 150 iterations. After 150 iterations, accuracy stabilized, and the loss remained very low, with no signs of overfitting. The model performs well in multimodal emotion recognition, achieving high accuracy and low loss without requiring extensive training.



(a):Accuracy training graph



(b):Loss function training

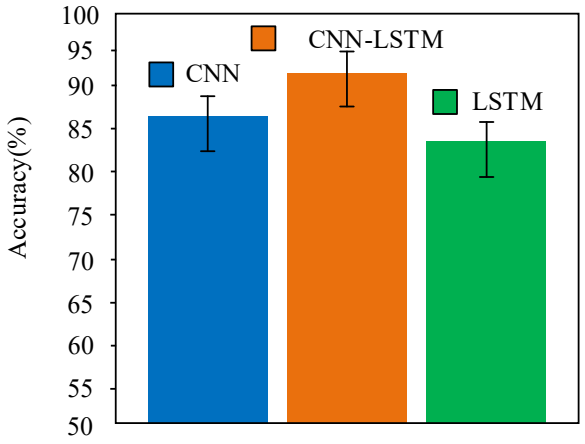
Figure 4. Training Results of the CNN-LSTM Model

B. Model comparison

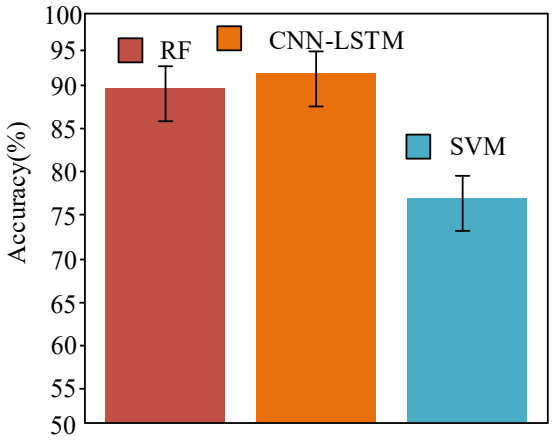
Figure 5 shows the multimodal emotion classification accuracy of various models, where the CNN-LSTM network model achieves an accuracy of 92%, the CNN model alone

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achieves 86.1%, and the LSTM model alone achieves 84.2%. Compared to the use of individual models, CNN-LSTM has more advantages. The accuracy of this emotion recognition was compared with that of other researchers. Among them, SVM classification accuracy is 89.3%, and RF classification accuracy is 77.1%. This model also greatly improves efficiency compared to traditional machine learning algorithms. n summary, the CNN-LSTM-based EEG emotion recognition method proposed in this paper has achieved good results in both recognition efficiency and accuracy.



(a) The accuracy of independent models



(b) The accuracy of traditional models

Figure 5. Model comparison

IV. Conclusion

This study proposes a CNN-LSTM model for recognizing emotions from EEG signals. CNN automatically extracts channel features, while LSTM effectively models dynamic temporal information. These features are integrated for multimodal emotion classification using softmax activation. This approach avoids the laborious processes of manually extracting, selecting, and reducing dimensionality of features in traditional machine learning methods, thereby saving time and enhancing efficiency. Compared to individual CNN and LSTM models, there is a significant increase in accuracy. The findings demonstrate the effectiveness of the CNN-LSTM network in modeling the spatiotemporal characteristics of EEG emotional features and achieving high accuracy. Although this study has improved efficiency compared to machine learning, the network model remains more complex than single-structure models, and current model complexity continues to be a focus for advancing multimodal emotion recognition. Future work will aim to extract high-dimensional features such as frequency-domain and time-frequency domain features, and integrate key electrode selection and simplify channel extraction to further enhance recognition efficiency.

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